

# VINAY KUMAR

Dallas, TX

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## Summary

Data Scientist with a Master's in Econometrics and Quantitative Economics, experienced in developing predictive models and optimizing workflows across healthcare and economic research. Proficient in Python and key libraries such as NumPy, Pandas, and Scikit-learn, with a strong grounding in statistical modeling. Skilled at translating complex data challenges into actionable insights while collaborating effectively with technical and non-technical stakeholders.

## EDUCATION

### The University of Texas at Dallas

May 2025

*Master of Science, Econometrics and Quantitative Economics*

- **Coursework:** Data Collection, Data Visualization, International Finance, Monetary Economics & International Banking

### The University of Delhi

May 2022

*Bachelor of Arts, Economics and Operational Research*

## WORK EXPERIENCE

### M&B Sciences Inc. | Data Scientist Intern

Jun 2024 - Present

- Mapped 10,000+ medical professionals to optimize resource allocation and support strategic decision-making, translating complex data into actionable insights for both technical and non-technical stakeholders.
- Enhanced forecasting accuracy by 20% through the development of multilevel regression statistical models in R Programming, addressing real-world data inconsistencies and market risk assessment.
- Optimized SQL queries to clean, transform, and integrate 70,000+ healthcare records, reducing data processing time by 50% and ensuring high-quality, scalable data foundations.
- Managed multiple high-priority projects and applied data-driven strategic planning, contributing to a 25% increase in business growth and improved client satisfaction.
- Produced comprehensive reports and presentations to clarify complex data analyses, effectively communicating technical decisions and insights to diverse audiences.

### Stat Skew | Data Scientist Intern

Mar 2023 - May 2023

- Digitized records and automated data entry processes with Python, reducing overhead costs by 15% annually and enhancing real-time access to critical information.
- Facilitated cross-functional collaboration to resolve key operational bottlenecks, analyzing workflow inefficiencies and providing actionable insights for improved process scalability.
- Designed and implemented Python-based ETL data pipelines to extract, transform, and load large datasets, reducing processing time by 21% while ensuring robust data integration and model readiness.

## PROJECTS

### Migration Trends in the United States

Sep 2024 - Dec 2024

- Proposed policies to boost immigration engagement by analyzing the Top 10 migrant inflow states in the U.S, identifying key socioeconomic factors that influence migration patterns, and offering insights to guide policymakers' decisions.
- Demonstrated a 9% to 3% drop in unemployment in Georgia and an 800,000+ increase in Florida's immigrant population, highlighting their financial contributions and economic impact through pivot tables.

### Texas Real Estate Market Analysis

Feb 2024 - May 2024

- Achieved a 25% increase in housing valuations and raised forecast accuracy by 15% by conducting market research and predictive financial analysis, utilizing linear regression, t-tests, factor analysis, and k-fold validation to ensure precision.
- Enhanced real estate trend insights by developing Tableau visualizations, utilizing correlation analysis, and time series evaluation to support data-driven decision-making.

### COVID-19 Analysis

Jun 2023 - Jul 2023

- Strengthened targeted interventions and proposed ways to reduce mortality rates in high-risk areas by 20% through in-depth analysis of global and regional COVID-19 trends, identifying infection hotspots and mortality patterns.
- Increased data analysis efficiency by 25% by enhancing vaccination data storage, refining data visualization in Power BI, and filtering out irrelevant records in SQL to ensure accuracy and streamline decision-making.

## KEY SKILLS

- **Languages:** Python(NumPy, Pandas), R, SQL, Power BI, Tableau, Microsoft Office (Excel, Word, PowerPoint, Outlook)
- **Data Science:** Predictive Modeling, Regression, Statistical Analysis, Machine Learning, NLP, Random Forest, EDA, K-means
- **Specializations:** Quantitative Analysis, Economic Research, Data Visualization, Risk Management, Data Modeling
- **Additional Skills:** Report Generation, A/B Testing, Hypothesis Testing, KPIs, Statistics, Entrepreneurial Mindset
- **Certification:** Google Data Analytics, Advanced SQL, R Programming, Excel VBA & Macros, Data Science